

Homework 6

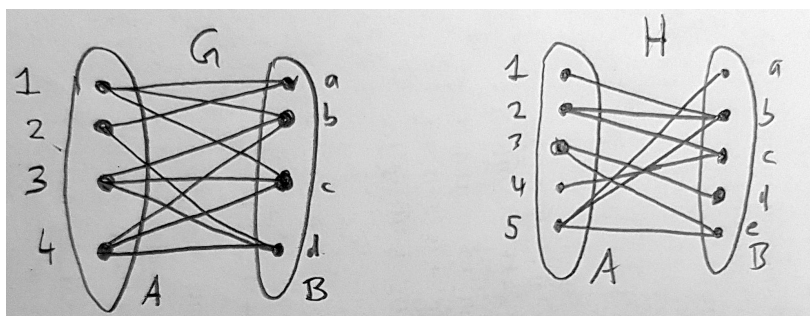
Instructor: Prof. Warnke

Due: Monday 05.11.2026, 9:00am

Please upload your solutions to Gradescope. Unless specifically told otherwise, you should always prove your assertions; sometimes, this may take the form of a list or a computation, but some form of justification will always be required. If submitting hand-written solutions, write all solutions clearly and very neatly; else, type them up using e.g. LaTeX (you cannot get credit for solutions your graders cannot read).

Problem 1 (15pts)

(a) In the following graphs G, H : either exhibit a matching of size $|A|$, or show that no such matching exists.



(b) In a probabilistic combinatorics graduate class there are 4 students (Laura, Mike, Paul, Tina), and 4 projects (Martingales, Phase Transition, Random Graphs, Thresholds). The project-preferences are Laura: M, PT; Mike: M, PT, T; Paul: PT, RG; Tina: PT, RG, T. Model this situation as a bipartite graph, verify Hall's condition, and find an assignment of students to distinct projects they are willing to work on.

(c) A school with 20 professors forms 10 committees, each containing 6 professors, such that every professor is on exactly 3 committees. Prove that it is possible to select a distinct representative from each committee.

Problem 2 (20pts)

(a) Prove that any k -regular bipartite graph $G = (A \cup B, E)$ on $n \geq 2$ vertices (with $k \geq 1$) has the following property: its edge set E can be written as the union of k edge-disjoint perfect matchings.

(Remark: in class we proved that such a k -regular bipartite graph contains a perfect matching; please re-prove this result, too.)

(b) Let $G = (A \cup B, E)$ be a bipartite graph with $|A| \leq |B|$, where all vertices $v \in A$ have exactly $a > 0$ neighbors and all vertices $w \in B$ have exactly $b > 0$ neighbors. Prove that G contains a matching of size $|A|$.

(Hint: first show that $a \geq b$ holds.)

Problem 3 (20pts)

(a) Let $G = (A \cup B, E)$ be a bipartite graph where $|N(X)| > |X|$ for every non-empty $X \subsetneq A$. Prove that every edge of G belongs to some matching M that covers A .

(Hint: use an idea from the proofs of Hall's theorem.)

(b) Let $G = (A \cup B, E)$ be a bipartite graph. Prove that the maximum matching of G has size $|A| - d^*$, where in $d^* := \max_S \{|S| - |N_G(S)|\}$ the maximum is taken over all vertex-subsets $S \subseteq A$.

(Remark: as part of this problem, please also re-prove the following result from class: if $|N_G(X)| \geq |X| - d$ for all vertex-subsets $X \subseteq A$, then G contains a matching M of size $|M| \geq |A| - d$ that covers all but at most d exceptional vertices of A .)

(c) Let $G = (A \cup B, E)$ be a bipartite graph on $2n$ vertices with $|A| = |B| = n$, and suppose that G does not have a perfect matching. Intuitively, your task is to establish that 'one failure of Hall's Condition implies two failures of Hall's Condition'. More concretely, your task is to show that there exist *two* distinct vertex subsets $X_1, X_2 \subseteq A \cup B$ with (i) $|X_i| > |N(X_i)|$, and (ii) $X_i \subseteq A$ or $X_i \subseteq B$.

(Hint: Given a set $S \subseteq A$ that violates Hall's condition, consider the set of vertices in B not adjacent to S .)